

PERSONALIZED VEHICULAR HEALTH DIAGNOSIS VIA LARGE LANGUAGE MODEL

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ABSTRACT

In the past decade, vehicle on-board diagnostics (OBD) systems have significantly improved by leveraging numerous electronic control units (ECUs) to monitor a wide array of real-time sensor data and report faults. However, traditional OBD systems often confuse drivers with generic warning lights that provide no context on issue severity. Recently, large language models (LLMs) have demonstrated advanced capabilities in knowledge acquisition and human-centered interaction, offering a promising solution for personalizing warning contents within existing OBD systems. In this paper, we propose an LLM-based OBD (LLM-OBD) system, an intelligent agent designed to deliver not only precise diagnostic information but also personalized guidance when a vehicle fault occurs. Moreover, the proposed LLM-OBD system can be used in coverage-challenged scenarios, such as highway, maritime routes and disaster situations, where terrestrial infrastructure is unavailable or unreliable. In these cases, non-terrestrial networks (NTN) can be leveraged to maintain reliable diagnostic capabilities. Specifically, LLM-OBD deploys a lightweight LLM on the vehicle to parse complex sensor data and driver personalization profiles into a concise query with negligible transmission overhead. Upon receiving the query through the satellite links, a responding LLM hosted on a central server generates accurate warning messages tailored to individual drivers, which are then relayed to the vehicle via the NTN connection. Experimental results demonstrate that the proposed LLM-OBD system reliably delivers personalized guidance to diverse drivers.

INTRODUCTION

Modern automobiles have evolved dramatically from purely mechanical machines into highly sophisticated systems integrating numerous electronic control units (ECUs). These ECUs manage a wide array of vehicle functions by continuously monitoring real-time sensor data, including engine temperature, wheel speed, tire pressure, throttle position, fuel injection parameters, etc. Utilizing

the data stream provided by these ECUs, on-board diagnostics (OBD) systems are designed to monitor vehicle health and promptly report faults in various automotive components. In recent years, improvements in sensor technology, on-board diagnostics, and vehicle communication protocols have allowed OBD systems to detect mechanical and electrical issues quickly and accurately. These systems collect data from different vehicle parts and identify unusual patterns indicating wear, malfunction, or failure. For example, acoustic signal processing was used in [1] to detect mechanical faults by analyzing data from microphones and vibration sensors. Similarly, the work in [2] modeled the complex relationships between multiple sensor signals using uncorrelated multilinear discriminant analysis. Recent research has also explored deep learning methods. A deep auto-encoder was used in [3] to detect mechanical faults from vibration signals during machining, however, the method is inefficient due to the low-dimensional data. To handle higher-dimensional data effectively, authors in [4] transformed one-dimensional motor current signals into two-dimensional images. These images were then analyzed using convolutional neural networks (CNNs), resulting in more accurate fault diagnosis.

Although OBD systems with the aforementioned techniques have been widely deployed in commercial vehicles, they face significant limitations in meeting drivers' demands. For example, existing OBD systems operate in a simplistic binary manner, using basic warning lights that only signal the presence of mechanical or electronic faults and provide limited guidance for addressing them. Such warning lights often cause confusion or unnecessary anxiety, as drivers struggle to judge the severity or urgency of the indicated problems without expert evaluation. For instance, a driver with limited mechanical knowledge might primarily need clear guidance on whether it is safe to continue driving on the highway when a warning indicator lights up unexpectedly. Conversely, an experienced automotive technician or mechanic would prefer detailed

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technical insights into specific issues, such as abnormal transmission noises, to assist in precise diagnosis and effective repair.

RELATED WORKS

To address these limitations, an intelligent and personalized OBD system becomes essential. Such the system should not only identify critical aspects of vehicle health but also deliver tailored warning contents and guidance according to driver profiles, including driving experience, mechanical knowledge, and personal preferences (e.g., the capability of lifting heavy tires). For example, instead of triggering a generic check-engine light, a more driver-understandable message, such as “Please tighten the fuel filling cap to avoid engine light,” should be presented to an inexperienced driver. Recent studies have demonstrated that large language models (LLMs) are highly capable of providing such message due to their capabilities of one-shot learning [5], [6] and human-centric interaction [7]. In vehicular settings, researchers have combined LLMs with image, video, and audio data to produce personalized driving-related guidance in autonomous vehicles [8], [9]. Additionally, LLMs have been widely utilized in fault detection and diagnosis for industrial machinery [10], [11]. For instance, research in [12] integrated aviation knowledge graph embeddings into an LLM to generate insightful and tailored guidance for different engineers, significantly improving the efficiency of aviation fault localization during aviation assembly and maintenance. Similarly, in the event of an unexpected vehicle mechanical fault, personalized warnings and guidance for drivers are crucial to effectively address the fault and, more importantly, mitigate drivers’ anxiety.

MOTIVATION AND CONTRIBUTION

However, in the field of vehicle OBD systems, the one-shot capability and human-centric collaboration advantages of LLM have not been exploited to intelligently determine vehicular health warning contents and provide proper guidance for individual drivers. Moreover, despite the significant

advancements in LLMs, existing literature often overlooks the feasibility of deploying these systems in coverage-challenged scenarios, such as remote highways, maritime routes, and disaster areas, where terrestrial infrastructure is often unavailable or unreliable. Therefore, to address these limitations and enhance driver confidence in vehicle reliability, this paper proposes a practical LLM-based OBD (LLM-OBD) framework enabled by non-terrestrial networks (NTN). To the best of our knowledge, this is the first study to integrate LLM-driven personalization into OBD systems while explicitly addressing the persistent coverage gaps in terrestrial vehicle-to-everything networks. Fig. 1 shows a comparison between the traditional OBD system and the proposed LLM-OBD system. The contributions are summarized as follows:

- We propose an LLM-OBD framework capable of operating in coverage-challenged environments through a distributed encoder-decoder architecture. To address the constraints of on-board computation and communication, a lightweight vehicle-side LLM functions as the semantic encoder, compressing high-dimensional sensor data and driver profiles into concise queries with negligible transmission overhead. Given the latency-tolerant nature of vehicle diagnostics, these queries are transmitted via satellite relays when terrestrial infrastructure is unavailable. Subsequently, a large-scale server-side LLM acts as the decoder, reconstructing the context to generate personalized diagnostic guidance, which is then relayed back to the vehicle via available terrestrial or non-terrestrial links.
- We introduce a multi-orbit NTN integration strategy capable of extending diagnostic connectivity to globally disconnected regions through the utilization of low, medium, and geostationary earth orbits (LEO/MEO/GEO). For coverage-challenged scenarios, the lightweight communication profile of LLM-OBD enables orbit-specific optimization by leveraging LEO’s

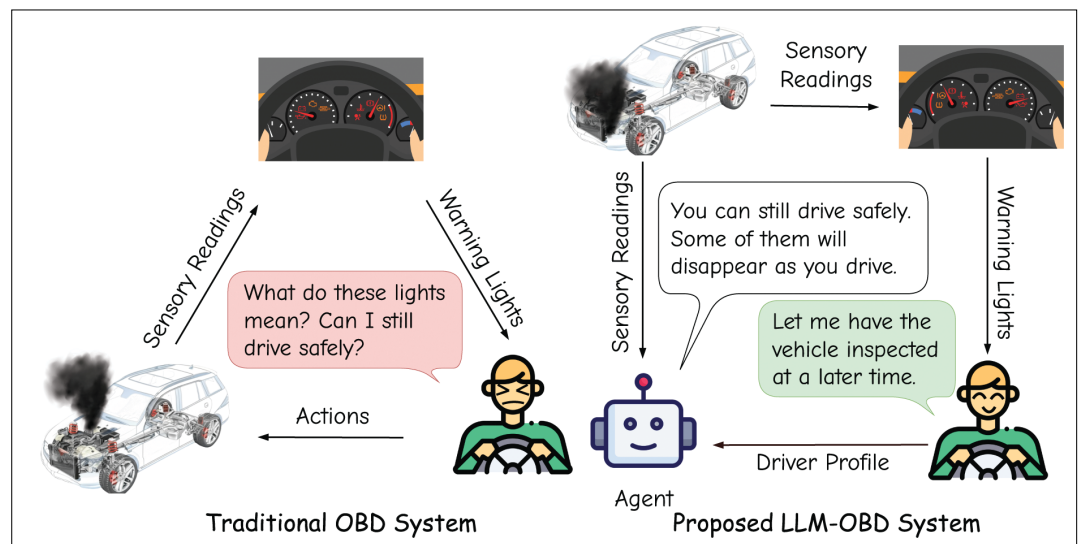


FIGURE 1. A comparison between a traditional OBD system and the proposed LLM-OBD system.

rapid signal for emergency response, utilizing MEO's narrowband links for remote highways, and exploiting GEO's wide-area stability for maritime logistics. Given the system's resilience to latency and handover interruptions, this design ensures that personalized vehicle fault diagnosis remains accessible regardless of the available satellites.

- We develop a knowledge-driven approach via retrieval-augmented generation (RAG) to improve the quality of the warning contents. Specifically, a dedicated knowledge-retrieval module ranks and extracts context relevant to the on-board LLM's query. The cloud LLM then leverages the retrieved contextual information to produce more driver-appropriate diagnostic content and recommendations. To enhance the quality of the retrieved knowledge, we further propose a query augmentation method to incorporate useful and related semantics of historical queries from other vehicles. Consequently, the relevance of the retrieved knowledge from the RAG knowledge base can be significantly improved.
- We develop a prototype of the proposed LLM-OB system on a low-power NVIDIA Jetson Nano to evaluate its performance in environments constrained by both on-board computation and connectivity. The experimental results demonstrate that the system generates personalized assistance strategies with minimal data overhead, validating the feasibility of NTN integration over bandwidth-limited satellite links. By coupling a lightweight on-board LLM with efficiently compressed transmission queries, the framework ensures deployment viability for a wide range of vehicles, including traditional fuel-based models lacking both powerful processors and high-gain terrestrial antennas.

METHODOLOGY

OVERVIEW

The LLM-OB system interacts with sensor data and human drivers to provide personalized on-board vehicular diagnostic information. However, the application of LLMs to domain-specific or knowledge edge-intensive tasks outside of their training data may result in erroneous outputs, commonly referred to as hallucinations. Although model fine-tuning can reduce the effect of hallucination by adapting the LLM to a specific domain, it is often computationally expensive, data-hungry, and impractical for resource-constrained vehicular environments. As a result, an alternative approach, RAG, is proposed as a way to mitigate this challenge by supplementing the LLM with external knowledge in real time. By incorporating RAG, the proposed LLM-OB system can generate more accurate and context-aware responses without requiring retraining. As shown in Fig. 2, the proposed LLM-OB system consists of four modules: on-board memory, knowledge retrieval, action, and feedback modules. Unlike existing LLM-based diagnostics in aviation or industrial machinery [10], [11], [12], which primarily target professional engineers with specialized technical backgrounds, the proposed system focuses on bridging the cognitive gap between complex sensor readings and the diverse technical proficiency of everyday drivers, translating raw fault codes into context-aware, actionable guidance for non-experts.

ON-BOARD MEMORY MODULE

The on-board memory module within the LLM-OB system is designed for data collection and query generation. For data collection, two distinct types of data, including long-term data and short-term data, are gathered to support vehicle health queries generation by the on-board lightweight LLM. Short-term data comprises temporary and real-time information collected

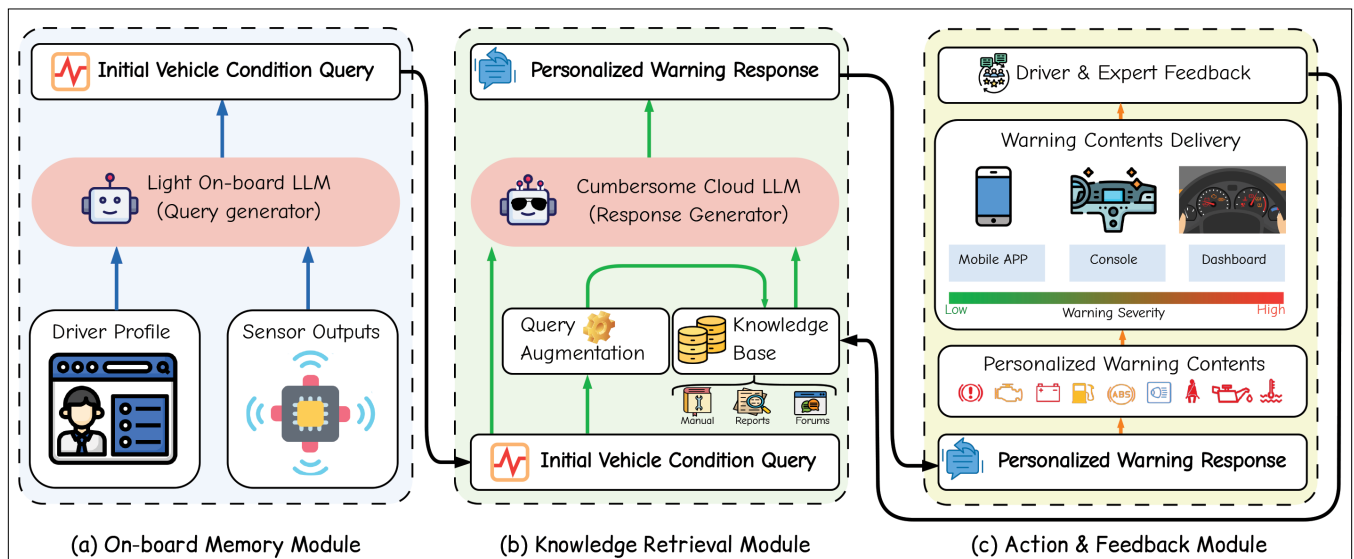


FIGURE 2. Workflow and taxonomy of our proposed LLM-OB system: a) The on-board memory module captures raw vehicle sensor data and provides it to the lightweight on-board LLM, which then generates a vehicle health query; b) The knowledge retrieval module retrieves relevant knowledge from RAG for the incoming query and produces personalized response. c) The action module determines how warning messages are conveyed to the driver while the feedback module enables the proposed LLM-OB system to continuously learn from its past experience.

during vehicle operation, including sensor outputs, vehicle location, and traveled distance. In contrast, long-term data involves the driver's personalized profile and vehicle health records, capturing persistent characteristics such as driver experience, maintenance knowledge, and historical vehicle conditions. To ensure practical feasibility, these profiles are constructed through a hybrid data pipeline. Initially, a "cold-start" profile is generated via a self-reported questionnaire on the vehicle's infotainment system or companion app, collecting explicit data such as the driver's age and self-assessed technical proficiency. Subsequently, this profile is dynamically refined by aggregating historical driving data (e.g., acceleration patterns indicating cautious versus aggressive driving) and integrating manufacturer service records. For query generation, a lightweight LLM (e.g., gemma3 1B) operates directly on the vehicle's computational unit, transforming both long-term and short-term data into retrieval queries. Specifically, the on-board LLM examines vehicular health indicators (e.g., engine temperature) and driver-related features (e.g., driver's knowledge level and driving patterns). Based on this analysis, initial vehicle health queries are formulated. For instance, abnormal engine temperature and pressure could suggest potential engine issues, while limited driving experience and erratic accelerator usage indicate inadequate driver knowledge and skills. Consequently, an initial query, e.g., **Error Type: Engine; Driver Experience: Inadequate. How do new drivers handle engine malfunctions on highways?,** can be generated. Note that **Error Type** and **Driver Experience** are predefined prefixes to enhance reasoning efficiency at the server. Since vehicles upload only concise and locally generated queries to the server, the uplink communication overhead of the vehicle can be significantly reduced compared with uploading raw data to the server in constrained wireless networks. By minimizing data throughput, the system enables vehicles in coverage-challenged scenarios, such as remote highways, maritime routes, or disaster zones, to maintain diagnostic connectivity via limited-bandwidth when terrestrial infrastructure is unavailable. Finally, these queries are then transferred to the knowledge retrieval module to retrieve relevant knowledge from the RAG knowledge database.

KNOWLEDGE RETRIEVAL MODULE

The knowledge retrieval module receives and processes the vehicle health query from the on-board memory module and then use them to retrieve relevant knowledge from RAG. In general, RAG uses the language embedding of the query to retrieve relevant elements and generates augmented outputs based on these elements. However, the retrieved knowledge can sometimes be naive, misaligned or irrelevant when the vehicle health query does not fully, or sometimes fails to, describe the vehicular sensor outputs and the driver's preferences. Due to limited computation resources on vehicles, the on-board LLM is subject to model compression via quantization, pruning, or weight sharing. This leads to performance drop of the on-board LLM in describing the short-term and long-term data. Therefore, to

address this issue, a query augmentation mechanism is proposed at the server side to precisely retrieve relevant knowledge from the RAG and, as a result, improve the performance of the system.

The key idea of the mechanism is to leverage historical queries previously generated for other vehicles of the same make, model, and year, as well as for drivers with similar profiles. Often, historical queries from other vehicles share similarities with queries generated for the current vehicle, although these similarities are typically overlooked during the query generation process. Such similarities can provide significant value; for instance, a historical query that successfully retrieved information addressing a driver's concerns can be effectively reused when similar situations arise. However, existing approaches lack an efficient mechanism to facilitate the reuse of such beneficial queries. To bridge this gap, inspired by recent advances in attention mechanisms and matching networks [13], we introduce an attention-based query augmentation mechanism that incorporates the context and knowledge of relevant historical queries. Intuitively, this mechanism acts like a digital form of collaborative troubleshooting. When a vehicle reports a vague issue, the system automatically references historical queries from other vehicles that exhibited similar symptoms but resulted in successful diagnoses. By borrowing specific descriptive details from those resolved cases, the system enriches the current, potentially ambiguous query.

Specifically, as shown in Fig. 3 the initial query q_{init} from the vehicle is transformed into a language embedding using a trained linear projector. Subsequently, this language embedding is compared with embeddings of historical queries through cosine similarity, which quantifies the context relevance between them. However, due to high redundancy in natural language, two semantically different queries can have undesired high similarity scores. To address this issue, we use the predefined prefix of each query to bound the semantical meaning during the comparison process. Assume there are K queries with the same prefix as q_{init} , the final generated query for knowledge retrieval is a weighted sum of historical queries based on the similarity scores, which can be expressed as

$$q_{\text{final}} = \alpha q_{\text{init}} + (1 - \alpha) \sum_{k=1}^K z(q_k, q_{\text{init}}) q_k \quad (1)$$

where $\alpha \in [0, 1]$ is the importance weight of the aggregated historical query embeddings, q_k denotes the k -th query with same prefix, and $z(q_k, q_{\text{init}})$ is the attention mechanism to determine the query aggregation weight of q_k . For example, $z(q_k, q_{\text{init}})$ is a softmax over the cosine distance $c(\cdot, \cdot)$ of the embeddings $\hat{q}_{\text{init}}$ and $\hat{q}_k$ of q_{init} and q_k , i.e., $z(q_k, q_{\text{init}}) = e^{c(\hat{q}_{\text{init}}, \hat{q}_k)} / \sum_{j=1}^K e^{c(\hat{q}_{\text{init}}, \hat{q}_j)}$.

Finally, based on q_{final} , RAG retrieves the most relevant knowledge and selects the top M items. The knowledge, driver profile, and queries are combined and fed to the cloud LLM, which reasons and generates the final vehicle health warning or suggestions contents.

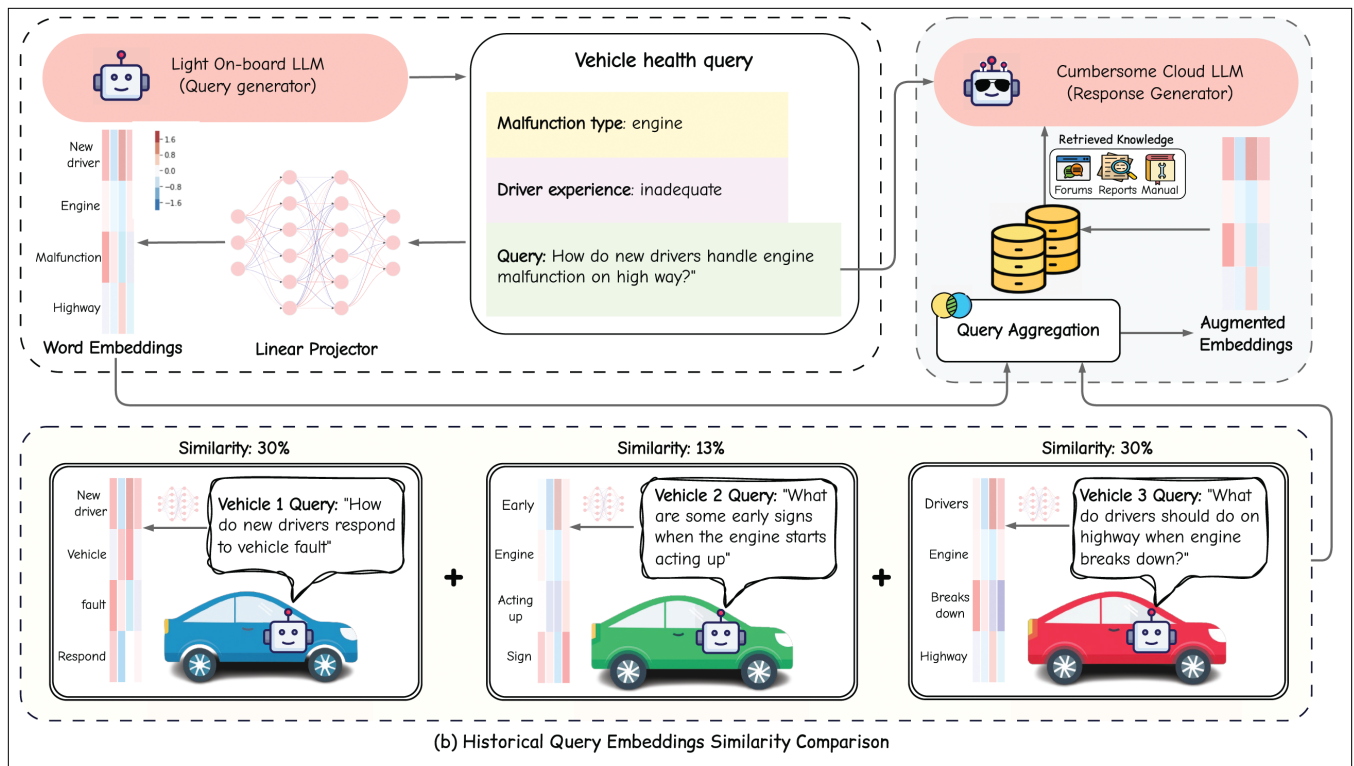


FIGURE 3. The attention-based query augmentation mechanism: a) Upon receiving the vehicle's query, the server feeds it into a linear projector to transform the text into vectorized word embeddings; b) Subsequently, the server runs similarity comparison between the transformed embedding and the embeddings of historical queries from other vehicle in terms of context relevance; c) The transformed embedding is augmented by incorporating relevant semantics from the embeddings of the historical queries.

ACTION MODULE AND FEEDBACK MODULE

The action module determines how warning messages are conveyed to the driver by selecting from three different delivery methods based on the severity of the situation. For critical warnings, such as engine overheating caused by coolant leakage, an urgent alert along with an audio alarm will immediately appear on the vehicle's dashboard in order to instruct the driver on the necessary actions. If the issue is less urgent but still requires driver attention, like an upcoming vehicle maintenance, the system displays a reminder on the dashboard at an appropriate time. Lastly, some messages may not indicate a fault but rather provide recommendations aimed at extending the lifespan of vehicle components. For instance, the driver is advised to avoid aggressive acceleration to prolong transmission life. Unlike critical alerts or reminders that are actively pushed to the driver, these advisory notes are passively made available as memos, which the driver can consult at their convenience.

The feedback module enables the proposed LLM-OBd system to continuously learn from its past experience while ensuring high reliability in safety-critical contexts. In this module, the key idea is to insert informative query-response pairs into the RAG knowledge database so they can be retrieved as contextual knowledge for future queries from vehicles. The updated RAG knowledge base is curated using high-integrity sources, including official equipment manufacturer manuals, verified service bulletins, and recall notices. To support continuous improvement, we introduce a two-layer human feedback mechanism for regularly

updating the RAG knowledge base. The first layer collects simple binary feedback from drivers (e.g., helpful or not helpful) regarding the usefulness of the response. Based on this, the second layer engages human professionals to review each query-response pair and assign a more nuanced score based on its relevance, technical accuracy, safety compliance, and clarity. Importantly, a higher score is not always assigned to those pairs marked as helpful by drivers in the first layer. This is because even responses that are deemed unhelpful by drivers may provide valuable negative examples, helping identify gaps in reasoning or misinterpretations that can serve as instructive references for future retrieval. Finally, the top-scoring query-response pairs, regardless of whether they were positive or negative examples, are selected for insertion into the RAG knowledge database.

INTEGRATION OF LLM-OBd INTO NTN

Satellites are pivotal for NTN to extend connectivity, covering broad geographical areas where terrestrial infrastructure is absent or too costly to establish. In this context, the proposed LLM-OBd framework is uniquely suited for NTN integration due to its data-efficient design. Unlike traditional diagnostic systems that require continuous high-bandwidth streaming of raw sensor readings, our system transmits only concise, preprocessed text queries with negligible transmission overhead. This lightweight communication profile allows the system to flexibly utilize different satellite orbits, including LEO, MEO, and GEO, to address diverse coverage-challenged scenarios.

LEO satellites operate at altitudes between 500 km and 2,000 km, LEO satellites offer low

latency and high signal strength, making them ideal for disaster situations where terrestrial networks are compromised. In such dynamic environments, where emergency response vehicles require rapid diagnostics, the low latency of LEO ensures timely interaction. Furthermore, although LEO satellites move rapidly relative to the ground, the small size of the generated health queries (typically a few kilobytes) ensures successful data packet delivery before handovers between satellites occur. MEO satellites are often used for navigation services, serve as an effective middle ground for vehicles traversing remote highways. These satellites offer a balance of stable coverage and moderate latency. Since our system decouples the heavy computational load to the cloud and retains only a lightweight query generator on the vehicle, it can potentially leverage existing MEO-based messaging links (e.g., search and rescue return links) to transmit diagnostic queries. This ensures that even traditional fuel-based models with basic satellite hardware can access personalized fault diagnosis without requiring the expensive, high-gain terminals typically needed for broadband data. Lastly, GEO satellites operate at approximately 36,000 km, providing wide-area, stable coverage essential for maritime routes. While GEO links introduce higher latency (approximately 600 ms), they are highly effective for the proposed system due to the “latency-tolerant nature” of vehicle diagnostics. Since the diagnostic guidance, such as advisory memos or maintenance reminders, does not require sub-millisecond real-time control, the system can effectively utilize GEO backhaul to deliver accurate, personalized warning content to vessels operating far beyond the reach of terrestrial towers.

EXPERIMENT

The successful implementation of the LLM-OB system relies on close collaboration among diverse stakeholders, including sensor data API developers, vehicle manufacturers, and cloud service providers. Such coordination is crucial for the development, integration, and deployment of an LLM-OB system capable of enhancing vehicular health management and providing drivers of all experience levels with clear, actionable guidance when vehicle issues arise.

PROTOTYPE SETUP

As shown in Fig. 4, we developed a prototype comprising two main components: a resource-constrained on-board computing unit, implemented on an NVIDIA Jetson Nano edge device, and a high-capacity cloud server equipped with three NVIDIA RTX 3090 GPUs. Using *llama.cpp*¹ as the deployment framework for both platforms, the cloud server hosts a LLaMA model with 30 billion parameters to serve as the response generator, while the vehicle unit runs a lightweight Gemma3 model with 1 billion parameters as the query generator to accommodate its limited computational resources. To retrieve relevant information from the RAG knowledge base, we measure semantic similarity between the vehicle-generated health query and stored knowledge using BERTScore [14]. Specifically, we convert both the query from the on-board LLM and the textual entries in RAG into token embeddings, then apply BERT to compute token-level alignment, yielding an F1-score. A value closer to 1 indicates stronger relevance between the retrieved knowledge and the query. To ensure the retrieved content is not only semantically relevant but also safe and accurate, we implement a pre-ranking mechanism. Each entry in the RAG is assigned a reliability score derived from expert verification and historical user feedback. Consequently, the final retrieval ranking is determined by a weighted combination of the semantic BERTScore and the reliability score. Lastly, given that the transmission of vehicular health information is not latency-sensitive, the communication between the on-board computing unit and the cloud server can be considered to be lossless. Thus, we establish the connection between the on-board computing unit and the cloud server over a local network.

PERSONALIZATION GENERATION

By constantly identifying the key characteristics of the driver and vehicle, the on-board query generator analyzes how those factors affect vehicle health. In this paper, we consider three sample drivers with different backgrounds as follows:

- Driver 1 - George (60 years old, male, with a full license and living in Toronto): An

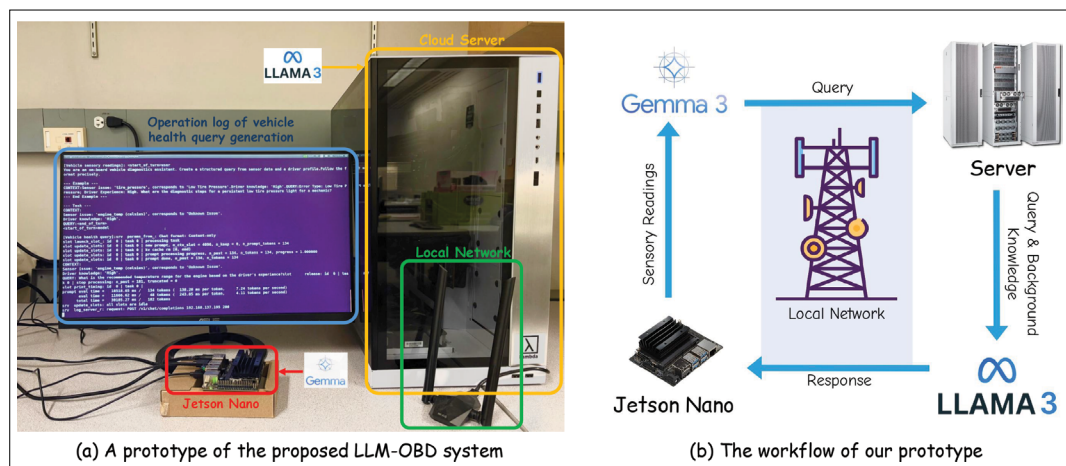


FIGURE 4. Experimental setup. a) A Jetson Nano serves as the on-board edge compute unit; its low-power profile enables deployment across vehicle types, including fuel-based vehicles. b) The vehicle uploads health queries synthesized from sensor readings and driver profiles to the server, which returns personalized warnings and guidance based on the diagnostic results.

¹ <https://github.com/ggml-org/llama.cpp>

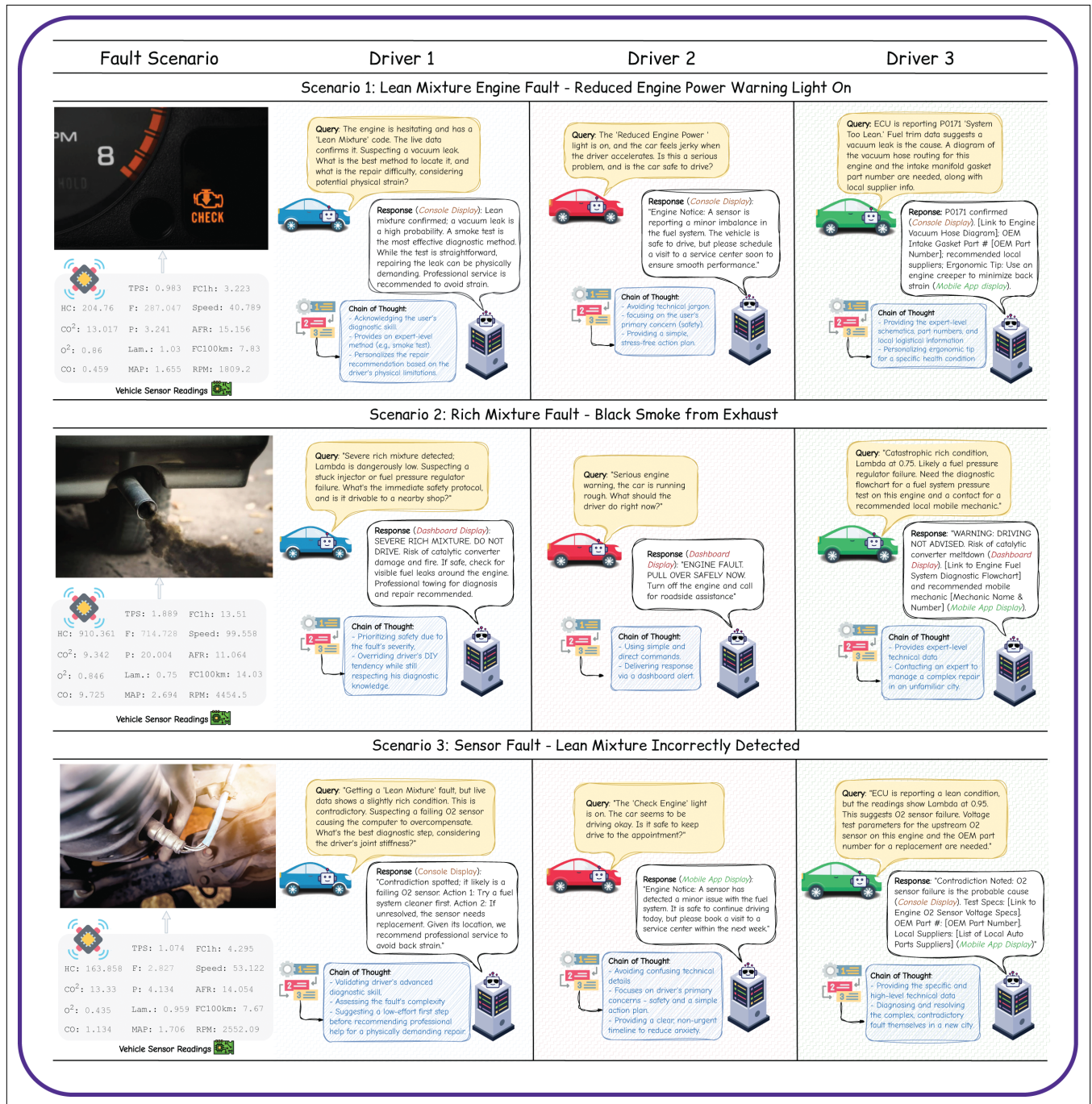


FIGURE 5. The inferences results of LLM-OBD in different scenarios from the driver profiles and EngineFaultDB dataset.

experienced driver, George is comfortable with basic maintenance. He can troubleshoot common mechanical issues and perform tasks such as changing engine oil and replacing air filters but he has difficulty in lifting heavy components.

- Driver 2 - Linda (24 years old, female, with an intermediate license and living in Ottawa): As a cautious and relatively new driver, Linda strictly adheres to the recommended service schedule. She prefers simple, routine tasks but tends to avoid complex inspections and struggles with using tools.
- Driver 3 - Chen (42 years old, male, with a full license and recently arrived in Canada):

A highly skilled individual, Chen can perform complex repairs, such as replacing a fuel pump or suspension struts. His work, however, is challenged by physical limitations, including severe myopia and a herniated disc.

By correlating the above driver-specific and their vehicle-specific features, the query generator produces an initial query that captures the most relevant aspects of the current operational state. For example, abnormal readings from the engine temperature and oil pressure sensors may be interpreted differently depending on the driver's maintenance experience and accident history. A novice driver with limited

Sensor	Description
MAP	Manifold Absolute Pressure: a measure of the pressure within the intake manifold, calculating the optimal air-fuel mixture for combustion.
TPS	Throttle Position Sensor: an electronic sensor that monitors the position of the throttle valve in a vehicle's engine
F	Force: Represents the engine's torque or rotational force.
P	Power: Quantifies the rate at which energy is transferred in the engine.
RPM	Revolutions Per Minute: Indicates the engine's crankshaft rotates per minute
FC1h	Fuel consumption 1/h: Illustrates the engine's fuel consumption rate.
FC100km	Fuel consumption 1/100km: Relays the engine's fuel efficiency over a given distance.
Speed	Speed (km/h): The vehicle's travel speed.
CO	Carbon monoxide: CO concentration in the exhaust gases.
HC	Hydrocarbons: Concentration of unburnt hydrocarbons in the exhaust.
CO ²	CO ₂ concentration in the exhaust, indicative of combustion efficiency.
O ²	Oxygen amount in the exhaust.
Lam	Lambda: The air-fuel equivalence ratio.
AFR	Air-Fuel Ratio: Ratio of air to fuel in the combustion chambers.

TABLE 1. Notations of Sensors.

technical knowledge might require simplified, safety-oriented guidance, whereas an experienced driver could benefit from detailed diagnostic insights to support self-maintenance. This continuous personalization process ensures that the generated queries reflect both the physical condition of the vehicle and the contextual background of the driver. As new sensory data and profile updates become available, the query generator dynamically refines the vehicle health query, enabling the server to generate accurate and personalized warning contents.

PERFORMANCE EVALUATION: CASE STUDY WITH ENGINE FAULT

The LLM-OBD system was tested using Engine-FaultDB [15] as the vehicle sensory dataset combining with 3 synthetic driver profiles. The agent's performance and robustness were evaluated using three different engine fault scenarios, including rich mixture, lean mixture, and sensor fault. A rich mixture fault indicates that there's too much fuel relative to air. Excessive fuel results in unburned hydrocarbons, causing black smoke to be expelled from the exhaust. A lean fuel mixture occurs when there is too much air or not enough fuel in the engine's combustion process. The engine may lack power during acceleration, and it might feel sluggish or hesitant. Lastly, sensor fault refers to a condition where the engine's electrical system, particularly the voltage supplied to critical components (e.g., sensors and actuators) drops below the expected operating range and causes incorrect readings.

Fig. 5 illustrates the on-board LLM's queries, the cloud LLM's responses, and the Chain-of-Thought (CoT) for three scenarios, along with the sensory readings listed in Table 1. Across these cases, the LLM-OBD system demonstrates its ability not only to detect mechanical faults but also to tailor warnings according to drivers' individual needs and preferences. In Scenario 1, when a "reduced engine power" warning occurs, the system adapts its response based on driver experience. The inexperienced driver receives a qualitative, safety-oriented message aimed at reassurance, while the expert driver is given a quantitative message containing diagnostic codes and part numbers. Since the fault is non-critical, the alert is delivered through the console display, ensuring the driver is informed without being distracted. In Scenario 2, a serious engine malfunction is detected. Here, the warning is issued directly on the dashboard display to ensure immediate visibility and rapid driver response. In Scenario 3, a sensor reading fault is identified. Drivers 1 and 3, who have sufficient technical knowledge, receive detailed quantitative guidance for self-repair. In contrast, driver 2 lacks the necessary background. For this driver, presenting a quantitative warning on the console might cause unnecessary anxiety. Instead, the system uses a mobile app display to provide simple, actionable guidance, such as advising a visit to a dealer.

PERFORMANCE ANALYSIS

The proposed LLM-OBD system acts as an adaptable diagnostic companion, functioning as an instructional advisor for novices or a technical expert for professionals. In our prototype, the end-to-end time for fault localization, query generation, and solution synthesis ranges from 1 to 2 minutes, depending on the complexity of the requested guidance. This processing duration confirms the latency-tolerant nature of the workflow, making it highly compatible with NTN. Since the computational inference time significantly exceeds the propagation delay of satellite links (even for high-orbit GEO satellites), the system can reliably deliver faster-than-manual diagnostics to drivers in remote or maritime environments, meeting practical safety needs without being hindered by the latency constraints of satellite backhaul. While the hybrid architecture maximizes computational efficiency, we acknowledge the challenges posed by connectivity loss and data privacy in real-world deployment. To mitigate the impact of intermittent network availability, the on-board module can be configured to cache frequently accessed query-response pairs, enabling the system to provide basic offline guidance for common faults. Furthermore, the vehicle's traditional binary warning system remains fully operational as a fail-safe, ensuring safety-critical alerts are never suppressed by connection failures. Regarding privacy and security, the proposed architecture inherently minimizes risk by transmitting only synthesized text queries rather than raw, high-resolution sensor readings.

CONCLUSION

In this article, we proposed LLM-OBD, an intelligent diagnostic agent designed to provide personalized guidance to drivers of varying experience levels. By utilizing a distributed architecture that decouples

on-board query generation from cloud-based response synthesis, the system minimizes uplink bandwidth requirements, making it uniquely compatible with NTN. This capability allows the system to extend its reach beyond terrestrial boundaries, utilizing satellite backhaul to serve vehicles in coverage-challenged scenarios. Our experimental evaluation demonstrates that the system is not only feasible on resource-constrained edge devices but also sufficiently delay-tolerant to operate reliably over diverse satellite orbits. Essentially, LLM-OBd provides a practical solution for intelligent and human-centric vehicle fault diagnosis, improving vehicle safety and reliability regardless of terrestrial infrastructure availability.

REFERENCES

- [1] A. Suman, C. Kumar, and P. Suman, "Early detection of mechanical malfunctions in vehicles using sound signal processing," *Appl. Acoust.*, vol. 188, Jan. 2022, Art. no. 108578.
- [2] Z. Xia et al., "Real-time fault detection and process control based on multi-channel sensor data fusion," *Int. J. Adv. Manuf. Technol.*, vol. 115, no. 3, pp. 795–806, Mar. 2021.
- [3] B. Luo et al., "Early fault detection of machine tools based on deep learning and dynamic identification," *IEEE Trans. Ind. Electron.*, vol. 66, no. 1, pp. 509–518, Jan. 2019.
- [4] Y. Dai et al., "Development of deep learning-based cooperative fault diagnosis method for multi-PMSM drive system in 4WID-EVs," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–13, 2024.
- [5] D. Rivkin et al., "AIoT smart home via autonomous LLM agents," *IEEE Internet Things J.*, vol. 12, no. 3, pp. 2458–2472, Feb. 2025.
- [6] H. Peng, C. Liu, and H. Li, "Large-language-model-enabled health management for internet of batteries in electric vehicles," *IEEE Internet Things J.*, vol. 12, no. 6, pp. 6082–6094, Mar. 2025.
- [7] W. Huang et al., "VoxPoser: Composable 3D value maps for robotic manipulation with language models," in *Proc. 7th Conf. Robot Learn.*, Nov. 2023, pp. 540–562.
- [8] Z. Xu et al., "DriveGPT4: Interpretable end-to-end autonomous driving via large language model," *IEEE Robot. Autom. Lett.*, vol. 9, no. 10, pp. 8186–8193, Oct. 2024.
- [9] H. Shao et al., "LMDrive: closed-loop end-to-end driving with large language models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 15120–15130.
- [10] Y. Wang et al., "RmGPT: A foundation model with generative pretrained transformer for fault diagnosis and prognosis in rotating machinery," *IEEE Internet Things J.*, vol. 12, no. 20, pp. 41562–41573, Oct. 2025.
- [11] Y. Liu et al., "Brain-like cognition-driven model factory for IIoT fault diagnosis by combining LLMs with small models," *IEEE Internet Things J.*, vol. 12, no. 16, pp. 32296–32309, Aug. 2025.
- [12] L. Peifeng et al., "Joint knowledge graph and large language model for fault diagnosis and its application in aviation assembly," *IEEE Trans. Ind. Informat.*, vol. 20, no. 6, pp. 8160–8169, Jun. 2024.
- [13] O. Vinyals et al., "Matching networks for one shot learning," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst. (NIPS)*, Dec. 2016, pp. 3637–3645.
- [14] T. Zhang et al., "BERTScore: Evaluating text generation with BERT," in *Proc. Int. Conf. Learn. Represent.*, 2020, pp. 1–10.
- [15] M. Vergara et al., "EngineFaultDB: A novel dataset for automotive engine fault classification and baseline results," *IEEE Access*, vol. 11, pp. 126155–126171, 2023.

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